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Faculty of Physics and Earth System Sciences
Bachelor of Science Physics

**Dynamical changes of polyethylenimine in porous silica
upon CO₂ sorption**

Bachelor Thesis

of

Luca Zarinelli

born 03.04.2003 in Hamburg

3798376

Supervisor: Dr. Martin Treß

Second Corrector: Prof. Dr. Frank Cichos

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Abstract

The reversible chemisorption of carbon dioxide by polyethylenimine (PEI) has drawn attention for its potential in molecular basket sorbents in which PEI is dispersed within mesoporous silica to enhance CO₂ uptake. While the chemical reaction between CO₂ and PEI is well understood, the molecular-scale dynamics remain largely unexplored. This study investigates those dynamics using Broadband Dielectric Spectroscopy (BDS), which probes the relaxation behavior of molecular dipoles across a broad frequency range. PEI-filled silica samples with 20 wt.% and 43 wt.% loadings are analyzed under alternating CO₂ and N₂ atmospheres at 30°C, 50°C, and 70°C. Three relaxation processes appear and are assigned to molecular motions of different scales. Temporal analysis reveals two layers of exponential relaxation. Desorption kinetics accelerate with increasing temperature, and unusual behavior emerges near 50°C during absorption, indicating complex underlying dynamics. The findings demonstrate the effectiveness of BDS in studying molecular dynamics of molecular basket sorbents.

Declaration

I declare truthfully that I have prepared the Bachelor thesis independently, that I have listed all the resources used completely and accurately and that I have marked everything that has been taken from other people's work unchanged or with modifications.

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Luca Zarinelli

Leipzig, April 10, 2025

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Contents

1	Introduction	1
2	Theory and Background	2
2.1	Theory of Dielectric Relaxation	2
2.1.1	Introduction	2
2.1.2	Molecular Dipoles	3
2.1.3	Debye Relaxation	4
2.2	Polyethylenimine and Molecular Basket	6
2.2.1	Chemisorption of CO ₂	6
2.2.2	Molecular Basket	7
3	Methods	8
3.1	Experiment	8
3.1.1	Samples	8
3.1.2	Setup	8
3.1.3	Measurement	9
3.1.4	Problems and Considerations	10
3.2	Data Analysis	11
3.2.1	Theoretical Models	11
3.2.2	Time Interpolation	13
3.2.3	Numerical Differentiation	14
3.2.4	Fitting	14
4	Results and Discussion	16
4.1	Desorption	17
4.2	Absorption	19
4.3	Exponential Relaxation Characterization	20
4.4	43-PEI	21
5	Conclusion and Outlook	23

List of Abbreviations and Symbols

BDS	Broadband Dielectric Spectroscopy / Spectrum
PEI	Polyethyleneimine
CO ₂ , N ₂	Carbon dioxide, nitrogen
CC	Cole/Cole model
Re	Real part operator
Im	Imaginary part operator
i	Imaginary unit
ε_0	Dielectric permittivity of vacuum: $\varepsilon_0 = 8.854 \cdot 10^{-12} As/Vm$
ω	Angular frequency of a periodic electric field
$\varepsilon^*(\omega)$	Complex dielectric function
$\varepsilon', \varepsilon''$	Real and imanginary parts of the complex dielectric function
ε'_{der}	Derivative (possibly numerical): $\varepsilon'_{der} = -\frac{\pi}{2} \frac{\partial \varepsilon'}{\partial \ln \omega}$
ε_S	Static dielectric constant $\varepsilon_S = \lim_{\omega \rightarrow 0} \varepsilon'(\omega)$
ε_∞	High frequency dielectric constant $\varepsilon_\infty = \lim_{\omega \rightarrow \infty} \varepsilon'(\omega)$
ω_P	Relaxation rate at maximal loss (ε''), peak parameter of the Cole/Cole model
$\Delta\varepsilon$	$\varepsilon_S - \varepsilon_\infty$, dielectric intensity parameter of the Cole/Cole model
α	Broadening parameter of the Cole/Cole model
$\omega_{Pk}, \Delta\varepsilon_k, \alpha_k$	Respective parameter of k-th term of a multi-CC fit
τ	Characteristic relaxation time of an exponential function
$[x]$	Array consisting of elements of type x

1 Introduction

The increasing concentration of atmospheric CO₂ due to anthropogenic emissions is a pressing global concern. While the problem can only be solved by reducing CO₂ emissions to zero, parallel direct air capture of the gas is indispensable. Much research is needed on this matter. In recent years, so-called molecular basket sorbents attracted strong scientific interest as a promising option for CO₂ capture. Molecular Baskets consist of a chemisorptive material that is spread on densely packed nanometer-scale pores. The pore structure improves gas diffusion and has high surface area for efficient use of the reactive material. Among the many candidate materials, polyethylenimine (PEI) has emerged as a versatile and chemically reactive polymer for selective CO₂ capture, owing to its high density of amine groups. While the chemical reaction is well understood and extensive research has been done on the properties of CO₂ uptake, open questions^[1] remain from the CO₂ uptake measurements that need insight into molecular scale dynamics. Recently, Tress et al.^[2] demonstrated the effectiveness of Broadband Dielectric Spectroscopy (BDS) to better understand polyethylenimine on the molecular scale. BDS is the analysis of a material based on its reaction to an oscillating electric field. The mobility of dipoles inside the material can be accessed by this method. This study aims to expand their research to PEI-induced mesoporous silica.

2 Theory and Background

2.1 Theory of Dielectric Relaxation

To understand how Broadband Dielectric Spectroscopy (BDS) can be used to study the dynamics of a material on the molecular scale, I will start with a short introduction of the underlying concept. This section is based solely on the book *Broadband Dielectric Spectroscopy* by Kremer and Schönhal^[3]. Demtröder^[4] and Fröhlich^[5] were also read for further understanding, but are not cited.

2.1.1 Introduction

Everything we need can be derived from Maxwell's equations.

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (2.1)$$

$$\nabla \times \mathbf{H} = \mathbf{j} + \frac{\partial \mathbf{D}}{\partial t} \quad (2.2)$$

$$\nabla \cdot \mathbf{D} = \rho_e \quad (2.3)$$

$$\nabla \cdot \mathbf{B} = 0 \quad (2.4)$$

Note that $\mathbf{E}$ and $\mathbf{H}$ denote the electric and magnetic fields, $\mathbf{D}$ the dielectric displacement, $\mathbf{B}$ the magnetic induction and $\mathbf{j}$ and ρ_e the current density and charge density. We are interested in the dielectric displacement for small electric fields:

$$\mathbf{D} = \varepsilon^* \varepsilon_0 \mathbf{E}, \quad (2.5)$$

where ε_0 is the vacuum permittivity and ε^* the complex dielectric function. In BDS, we search for time-dependent processes in the material, which cause ε^* to be time-dependent according to Maxwell's equations. For a periodic electric field $\mathbf{E}(t) = \mathbf{E}_0 e^{-i\omega t}$ it is defined as follows:

$$\varepsilon^*(\omega) := \varepsilon'(\omega) - i\varepsilon''(\omega), \quad (2.6)$$

where ω is the angular frequency of the electric field, and $i := \sqrt{-1}$ the imaginary unit. Since both $\mathbf{E}$ and $\mathbf{D}$ are vector quantities, the dielectric function ε^* is generally a tensor. However, for this study, we can assume isotropic materials, where it can be treated as a scalar function.

2.1.2 Molecular Dipoles

The part of the dielectric displacement caused by the material is called (macroscopic) polarization $\mathbf{P}$.

$$\mathbf{P} = \mathbf{D} - \mathbf{D}_0 = (\varepsilon^* - 1)\varepsilon_0\mathbf{E} \quad (2.7)$$

The polarization arises from microscopic dipole moments $\mathbf{p}_i$ in the material reacting to the external field.

$$\mathbf{P} = \frac{1}{V} \sum_i \mathbf{p}_i \quad (2.8)$$

V denotes the volume occupied by the material. Whereby a dipole moment can a priori be any volume Ω with shifted positive and negative electric centers of gravity. This shift can be permanent or induced by a local electric field $\mathbf{E}_{loc}$. Given an arbitrary charge distribution ρ_e on the volume, it is calculated by

$$\mathbf{p} = \int_{\Omega} \mathbf{r} \rho_e(\mathbf{r}) d^3r. \quad (2.9)$$

There can be various induced polarizations – such as displacements of atomic orbitals with respect to their associated nuclei – that occur at very fast rates, comfortably outside of the BDS frequency range. These are denoted by the constant $\mathbf{P}_{\infty}$. Inside the range are the time scales of movements of molecules or parts thereof. Assuming N molecular dipoles of the same kind, the sum (2.8) simplifies to

$$\mathbf{P} = \frac{N}{V} \langle \boldsymbol{\mu} \rangle + \mathbf{P}_{\infty}. \quad (2.10)$$

For different kinds of dipoles, a sum over all kinds is needed. To get access to the mean dipole moment $\langle \boldsymbol{\mu} \rangle$, we need to consider the distribution of their orientations. In the absence of interactions between dipoles and under the assumption $\mathbf{E}_{loc} = \mathbf{E}$, it can be shown that

$$\langle \boldsymbol{\mu} \rangle = \frac{\mu^2}{3k_B T} \mathbf{E} \quad (2.11)$$

using Boltzmann statistics. T is the Temperature and k_B the Boltzmann constant. Applying this to the terms (2.7) and (2.10) we obtain

$$\varepsilon_S - \varepsilon_{\infty} = \frac{1}{3\varepsilon_0} \frac{\mu^2}{k_B T} \frac{N}{V} \quad (2.12)$$

where $\varepsilon_S := \lim_{\omega \rightarrow 0} \varepsilon'(\omega)$ is the static permittivity and $\varepsilon_{\infty} := \lim_{\omega \rightarrow \infty} \varepsilon'(\omega)$ accounts for the high-frequency contributions. Without the assumptions of no interactions and a spatially constant electric field, (2.12) gets much more complicated.

2.1.3 Debye Relaxation

Simply put, dielectric relaxation describes the delayed reaction of the material to a change in the electric field. More accurately, it is described by linear response theory, assuming a time dependent disturbance $\mathbf{E}(t)$ with the response $\mathbf{P}(t)$.

$$\mathbf{P}(t) = \mathbf{P}_\infty + \varepsilon_0 \int_{-\infty}^t \varepsilon(t-t') \frac{d\mathbf{E}(t')}{dt'} dt' \quad (2.13)$$

$\varepsilon(t)$ is the time-dependent dielectric function, which is related to $\varepsilon^*(\omega)$ through Fourier transformation. In the simplest model, the polarization is assumed to pursue equilibrium exponentially with the characteristic Debye relaxation time τ_D :

$$\frac{d\mathbf{P}}{dt} = -\frac{1}{\tau_D} \mathbf{P}(t). \quad (2.14)$$

With this assumption, the Debye model for dielectric relaxation can be obtained:

$$\varepsilon^*(\omega) = \varepsilon_\infty + \frac{\Delta\varepsilon}{1 + i\omega\tau_D}, \quad (2.15)$$

where $\Delta\varepsilon := \varepsilon_S - \varepsilon_\infty$. For a rigorous derivation, one may read §10 of *Theory Of Dielectrics* by Herbert Fröhlich^[5].

Graphical takeaway

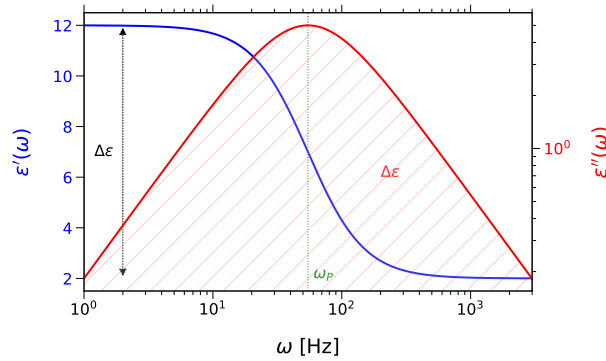


Fig. 2.1 Visualization of the Debye model and its parameters ω_P and $\Delta\varepsilon$

The real part ε' of the complex dielectric function is proportional to the energy stored reversibly in the system per period. Graphically speaking, the Debye model predicts it to make a step-like transition at frequency $\omega_P := \tau_D^{-1}$, because τ_D is the time scale at which the observed type of dipole responds to the electric field. At lower frequencies, all dipole alignments equilibrate, leading to a convergence of the stored energy for $\omega \rightarrow 0$. For higher frequencies, the movement of the dipoles is minimal and the only significant contribution is the induced polarization $\mathbf{P}_\infty$. The size of the step is the dielectric intensity $\Delta\varepsilon$. These two parameters can be accessed through BDS by measuring the permittivity of the material

of interest with respect to the frequency of the electric field. They give insight into the mobility of the observed type of dipole (ω_P) and to $\mu^2 N$ through (2.12).

The imaginary part, on the other hand, is proportional to the dissipated energy per period. It stores the same information as the real part as it is related through Kramers/Kronig. In the Debye model ε'' has a symmetric peak at $\omega = \omega_P$. Additionally, in the $\ln(\omega)$ plot, $\Delta\varepsilon$ describes the area under the function, and in the log-log plot, the shoulders of the peak fall linearly with slope 1.

Practical considerations

In practice, the Debye model is rarely observed. High concentrations of the dipole come with interactions between the dipoles and local electric fields distinct from $\mathbf{E}(\omega)$. These result in inhomogeneity uttering in broadening and asymmetry. The appropriate model for this study is discussed in methods, 3.2.1. In addition, if multiple kinds of dipoles are present, a sum of multiple relaxation terms is needed, analogous to the considerations for (2.10). In that case, ε' shows multiple steps.

From now on, only ω_P is used to discuss mobility, as τ will be used to discuss temporal evolutions of the material upon gas exposure.

2.2 Polyethylenimine and Molecular Basket

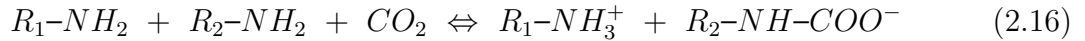
Polyethylenimine (PEI) is a polymer based on a simple repeating unit consisting of an amine group and two carbon aliphatic spacers.



Depending on the synthesis method, PEI can exist in either a linear or branched configuration. Linear PEI is composed exclusively of secondary amines, while branched PEI may contain primary, secondary, and tertiary amines. The classification of amine types refers to how many of the nitrogen's valence electrons are used in bonding with the polymer chain or other groups: a primary amine has one hydrogen replaced, secondary has two, and tertiary all three. Branched PEI is typically amorphous and liquid at room temperature. The high density of proton-accepting amine groups makes PEI particularly reactive with electrophilic species such as CO_2 . Importantly, this reaction is reversible under moderate conditions, allowing PEI to be regenerated and reused.

2.2.1 Chemisorption of CO_2

The CO_2 binding reaction requires two primary or secondary amines in order to form ammonium carbamate.



This reaction is thermally reversible. Additional reactions are possible under moist conditions, which can be ignored for this study.

Despite extensive studies on the changes in chemical composition, the molecular dynamics of PEI remain relatively unexplored. Recently, Tress et al.^[2] conducted dielectric analysis similar to this study on bulk and thin film PEI of very high molecular weight, demonstrating the high effectiveness of BDS to characterize polymer dynamics. Three distinct, asymmetric dielectric relaxation processes in the frequency range of 10^3 to 10^7 rad/s were observed. These processes are associated with different modes of polymer segmental motion. It was shown that CO_2 sorption slows down the dynamics substantially, indicating strong interactions between the ionic reaction products. More specifically, they postulate the ionic interactions of the reaction products ammonium carbamate and NH_3^+ to be substantially stronger than the hydrogen bonding of the amines, resulting in a significant reduction of chain mobility.

2.2.2 Molecular Basket

This section is based on [1, 2, 6–12]. Although PEI has favorable chemical properties for CO₂ binding, bulk PEI shows poor absorption performance in practice. Assuming the aforementioned strong cross-linking, this could be explained by slowing down of the outermost PEI layer after CO₂ sorption, hindering diffusion of the gas into deeper regions.^[2]

To overcome this limitation, the concept of the Molecular Basket was introduced by Xiaochun Xu et al.^[7] In this approach, PEI is dispersed as thin films within the nanometer-scale pores of a mesoporous material, typically silica-based supports such as SBA-15 or MCM-41. These frameworks provide large surface areas and fast gas diffusion across the whole system. Notably, mesoporous silica alone already exhibits slight physisorptive properties due to capillary condensation, with preferred selectivity for CO₂. When PEI is spread on the inner pore surfaces, the resulting hybrid material displays significantly enhanced CO₂ absorption rates and capacities, far exceeding those of bulk PEI. The inventors of the method even proposed a synergistic effect between physisorption within the pores and chemical absorption by PEI, causing absorption rates and capacities that cannot be explained by surface area and gas diffusion alone.

The kinetics of CO₂ uptake and release in PEI-induced mesoporous systems have been studied using various experimental methods. Those include gravimetric analysis^[12], which monitors sample mass changes under controlled temperatures and pressures, and real-time inlet and outlet CO₂ concentration measurements through infrared spectroscopy^[6]. Generally, exponential-like behavior is reported for both sorption and desorption processes, while different fitting models have been used. Hereby, the kinetics can be divided into a rapid part and a slower part, which is attributed to physisorption and chemisorption, respectively.

IR spectroscopy measurements by T. Beger^[1] using the same PEI-silica samples as this study revealed half-times of absorption and desorption kinetics. Unfortunately, the half-times were determined by the time taken to reach 50% of maximum capacity, not by fitting, which is problematic assuming two superposing processes and due to dependence on the maximum capacity. They observed half-times of several minutes for absorption. Desorption slows down significantly for low temperatures.^[1, 7, 12]

For atmosphere-like CO₂ concentrations, surprisingly fast CO₂ uptake and high capacity around 40 – 50 °C were reported for specific pore diameters and loadings, exceeding both lower and higher temperatures.^[1, 12] While Yihe Miao et al.^[12] propose possible macroscopic reasons, this study marks the start of a series of dielectric analyses on this matter. BDS could be a powerful framework to better understand the molecular dynamics.

3 Methods

3.1 Experiment

3.1.1 Samples

The experiments were conducted using the mesoporous silica powder *MB70-75/200* from CHROMATOREX^[13]. The grains have normally distributed diameters around 100 μm and isotropic pores of 7 nm diameter. With a pore volume of 0.77 ml/g, they are very suitable for PEI impregnation. The samples of PEI-induced silica were prepared by Tobias Beger^[1] from our Institute of Chemical Technology, Heterogeneous Catalysis, University of Leipzig. Preparation involved dissolving branched polyethylenimine of molecular weight of 600 g/mol in methanol, mixing the solution with the silica powder and finally drying out the methanol. PEI fillings of 20 and 42.9 percent by weight (*wt. %*) were used. The samples will be called *20-PEI* and *43-PEI*.

3.1.2 Setup

The dielectric analysis was carried out with an *Alpha-A Analyzer*^[14] from novocontrol connected to a parallel plate capacitor. A small amount of powder was placed on the lower capacitor plate, which was coated with a thin gold layer. A brass electrode was then placed on top, lightly pressed and wiggled to form a uniform, compact powder layer with minimal voids. The capacitor was housed in a small measurement cell designed to maintain a controlled atmosphere. The cell itself was submerged in nitrogen flow of a *Quatro Cryosystem*^[15], likewise from novocontrol, for temperature regulation. A PT100 sensor, attached to the outer brass cell wall, measured the temperature.

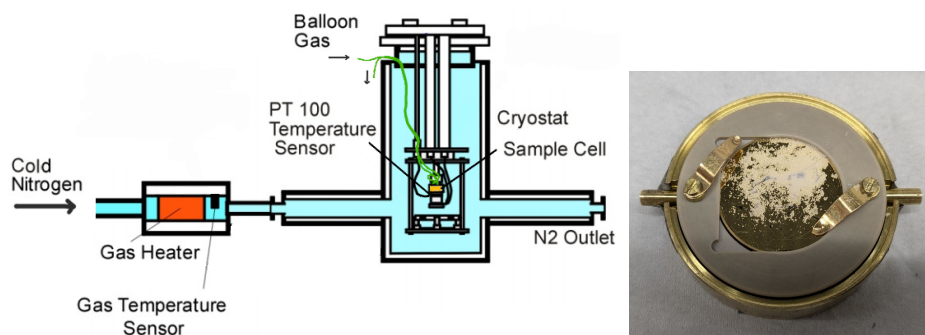


Fig. 3.1 (left) Sketch of the measurement cell attached to the balloon and inside the cryostat. Picture taken from novocontrol's website^[15] and edited. (right) Spread of the 43-PEI silica powder on the lower capacitor plate inside the measurement cell. The spread showed to not change significantly throughout the measurements for both fillings.

The measurement involved alternately exposing the capacitor to CO_2 and N_2 atmospheres. Therefore, gas was introduced via a balloon connected to the cell through a thin tube. A second tube allowed excess gas to escape, ensuring atmospheric control within the cell. The small inner volume of the cell facilitated rapid atmosphere exchange when switching balloons. The gas inside the balloon was transferred from pressurized gas cylinders.

3.1.3 Measurement

Dielectric spectra were recorded by applying alternating voltages at varying frequencies and measuring the real and imaginary parts of the permittivity. This “sweep” was continuously repeated, creating a time evolution of the spectrum. In order to observe temporal changes, one sweep should not take too long. On the other hand, we want to look at a wide range of frequencies at reasonable resolution. The compromise used in all measurements is a range from 10^{-1}Hz to 10^6Hz , with two points per decade, or rather one for low frequencies that take long to measure.

Preconditioning

Prior measurements with empty silica indicated the presence of initial strong polarization effects, likely due to residual moisture. These effects showed to disappear through short-term heating. Additionally, the carbon should be fully desorbed before the measurement.

Thus, the samples were preconditioned by heating them to 90°C at a nitrogen atmosphere, followed by an equilibration period. The temperature was then lowered to the desired temperature, with another equilibration phase before starting the measurement.

Measurement Sequence

The introduction of a new full gas balloon marks the start of (what will be called) a new *run*. Four to five directly consecutive runs make up one measurement of constant temperature. This measurement was conducted for the temperatures 30°C , 50°C and 70°C , for both PEI fillings.

The runs were sequentially numbered, starting with run 0, which corresponded to the first balloon after preconditioning. All plots are labeled according to the format $PP\text{-}TT\text{-}GG\text{-}j$, where PP represents the weight ratio of PEI filling, TT the measurement temperature, GG the gas, and j the run number.

Details

The preparation for 43-PEI was done without nitrogen, at air atmosphere. In return, the first measurement run was always a desorption run with nitrogen, whereas the 20-PEI measurements started with an absorption run. Lastly, the 30°C measurement of 20-PEI started with a nitrogen run and consisted of five runs. The run 20-50-CO₂-0 had an issue in the electric circuit and was left out.

3.1.4 Problems and Considerations

A notable potential source of error is that the gas temperature and pressure were not directly controlled.

Gas Temperature

The balloon was stored at room temperature, while the end of the tubing was partially inside the cryostat. However, due to the poor thermal conductivity of the plastic tubing, the temperature inside the tube's end is unlikely to match the cryostat temperature. While the brass cell wall ensured good heat conduction to the temperature sensor, a stable temperature gradient between the capacitor and the wall might still exist. This is fine with a constant intake gas temperature resulting in a lower - but still stable - actual temperature inside the capacitor. However, a varying intake gas temperature can affect kinetic measurements by introducing a temperature gradient that influences particle mobility. Some runs could have started with a slightly cooler balloon due to the prior transfer from a gas cylinder, with differing storing times at room temperature.

Gas Pressure

The partial gas pressure depends on the inflation of the balloon and potentially influences the CO₂ absorption rate. Hence, it is neither stable during a run, nor is the initial pressure perfectly consistent between runs. The initial balloon volume roughly ranged from 0.4 to 0.8 m³ and halved in about an hour. After that, the gas flow was very slow.

Gas Purity

When refilling the balloon, complete gas purity could not be ensured. However, the gas purity should have stayed above 98%, considering the volume ratio of the full balloons and the residual gas inside the balloon and tubing before filling. Inside the cell, gas purity may differ, because the cell was not perfectly gas-tight. Admixture of the cryostat atmosphere has to be considered at all times and could become significant as the balloon pressure decreases. The cryostat atmosphere is not assumed to be pure nitrogen, but rather a pollutant of unknown composition.

3.2 Data Analysis

All data processing and visualization was performed using Python, particularly the *NumPy*, *SciPy* and *Matplotlib* libraries. Some functions were developed with the assistance of the large language model *ChatGPT*^[16], but all implemented code was fully understood and validated. The whole code, as well as the raw data and extensive plots are accessible on GitHub^[17]. To explain the data processing in general terms, t , x and y may be used. t and x always refer to the time and frequency axes, while y stands for (possibly processed) measured data like ε' and ε'' .

3.2.1 Theoretical Models

Cole/Cole

To analyze the measured spectra, the Debye model (2.15) was extended by a broadening parameter α , which yields the Cole/Cole model^[3]:

$$\varepsilon_{CC}^*(\omega) = \varepsilon_\infty + \frac{\Delta\varepsilon}{1 + (i\omega/\omega_P)^\alpha} \quad (3.1)$$

Like in the Debye model, $\Delta\varepsilon$ and ω describe the relaxation strength and peak frequency, respectively, as discussed in 2.1.3. α accounts for heterogeneities that often appear for real materials. The value is constrained between 0 and 1 by its mathematical definition, where lower values imply a wider spread and 1 would imply a perfectly homogeneous Debye-like relaxation.

Since the spectral resolution was relatively low and due to severe superposition of several relaxation processes, distinctive conclusions about both relaxation shoulders could not be made. Thus, only the Cole/Cole model was used, assuming symmetric relaxation processes and focusing on relaxation strength and mean relaxation time. A sum of multiple Cole/Cole functions can be used to describe multiple relaxation processes.

Real Part Derivative

The individual relaxation processes and their characteristics are better isolated and more accessible in the loss spectrum compared to the spectrum of the real part^{[3][18]}:

- ω_P is uttered by a peak rather than a maximal slope
- $\Delta\varepsilon$ is uttered by an area rather than the before and after values. In the real spectrum it is hard to say where a process starts and ends, especially with multiple processes.

- The stretching parameter α is also more pronounced when looking at the high and low frequency shoulders of the loss spectrum rather than the broadening of a monotonically decreasing shape.

However, the real spectrum can be transformed into a loss-like form by calculating the derivative by $\ln(\omega)$ ^[3, 18]:

$$\varepsilon'_{der} := -\frac{\pi}{2} \frac{\partial \varepsilon'}{\partial \ln(\omega)} \quad (3.2)$$

In order to fit the spectrum in this form, the real part and derivative of (3.1) can be calculated analytically, as provided by Wübbenhorst and van Turnhout^[18]:

$$\begin{aligned} \varepsilon'_{der,CC} &= -\frac{\pi}{2} \frac{\partial}{\partial \ln \omega} \operatorname{Re}(\varepsilon_{CC}^*(\omega)) \\ &= \Delta\varepsilon \cdot \Omega \cdot A \cdot \frac{\cos(A - 2 \arctan(\frac{\Omega \sin A}{1 + \Omega \cos A}))}{1 + 2\Omega \cos A + \Omega^2}, \end{aligned} \quad (3.3)$$

where $A = \alpha \cdot \frac{\pi}{2}$ and $\Omega = (\omega/\omega_P)^\alpha$.

Exponential Decay

As shown in the Theory section (2.2), CO₂ uptake generally follows an exponential curve, while different models have been used. This group of models is justified by the fact that CO₂ absorption is limited by the number of available binding sites. As more sites are occupied, the absorption rate slows down. Underlying is the assumption of the slope being proportional to the current value:

$$\frac{dn}{dt}(t) = -\tau n(t) \quad (3.4)$$

Where in this specific case n would be the filled proportion of the capacity $n = \frac{N}{N_{max}}$ (with N the number of absorbed CO₂ molecules and N_{max} the maximum capacity) and $\frac{dn}{dt}(t)$ would be the CO₂ absorption rate at time t . It is expected that the molecular dynamics evolve at the same rates as CO₂ uptake as bound CO₂ should influence the molecular structure. To verify this, a simple exponential decay model was used to access the characteristic relaxation time $\tau > 0$.

$$x(t) = a \cdot \exp\left(-\frac{t}{\tau}\right) + c \quad (3.5)$$

Whereby $c > 0$ and $a \in \mathbb{R}$ are arbitrary offset and scaling values that were ignored. Another used model is the sum of two simple exponential functions, representing two processes of different characteristic times τ_1 and τ_2 . Note that $\tau_1, \tau_2, c > 0$ and $a, b \in \mathbb{R}$.

$$x(t) = a \cdot \exp\left(-\frac{t}{\tau_1}\right) + b \cdot \exp\left(-\frac{t}{\tau_2}\right) + c \quad (3.6)$$

3.2.2 Time Interpolation

As explained in 3.1.3, the whole BDS is not recorded at once, but rather by measuring one frequency at a time. A complete frequency sweep requires about 1.7 minutes, which is significant given that major changes in permittivity occurred within the first few minutes of each run.

To correct for this, an interpolation method was applied to align the data of all frequencies to a uniform set of “test” time points $[t_{int}]$. For every frequency and test time t_{int} the two neighboring raw data points were linearly interpolated to obtain an interpolated point $(t_{int}, y_{int}(\omega, t_{int}))$. With this approach, a three-dimensional dataset with uniform time and frequency axes was created for both the real and imaginary parts of the permittivity. Another benefit is that a higher resolution on the time axis is reached. Normally, linear interpolation does not add any real information, but in this case, in-between spectra are approximated. These consider the different movements of the single points, and thus contain real information. The spacing of the test time set was conservatively set to one minute.

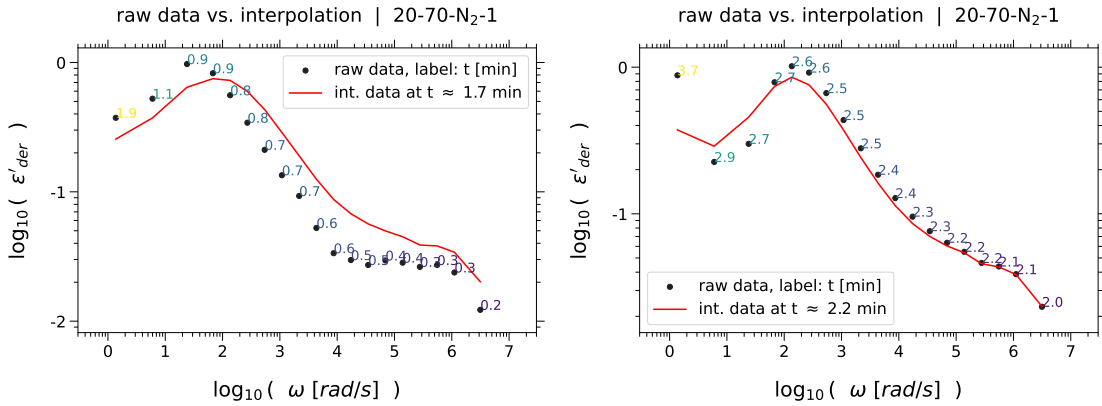


Fig. 3.2 Visualization of the interpolation on two examples (as labeled). The measurement times of the raw data spread over a large span resulting in a distorted spectrum compared to the interpolation.

Alternative interpolation techniques were considered, but ultimately rejected, including local linear regression and polynomial spline interpolation. Local linear regression is beneficial for reducing noise, which was not a major issue in the present dataset. However, for low-resolution data, it leads to a distortion of the slope. Polynomial spline interpolation, while being a better approximation for smooth data, can not be used on data that describes a non-differentiable shape. Specifically, the switching of gases resulted in an abrupt slope change and effectively non-continuous data, given the 1.7-minute spacing. Both methods can potentially lead to big errors here, making linear interpolation the most robust approach.

All subsequent data analysis and fitting procedures were based on the interpolated spectra.

3.2.3 Numerical Differentiation

After the interpolation process, the derivative ε'_{der} is calculated numerically from the real part through forward differentiation:

$$\varepsilon'_{der}(\omega_{der,j}) = -\frac{\pi}{2} \frac{\varepsilon'(\omega_{j+1}) - \varepsilon'(\omega_j)}{\ln \omega_{j+1} - \ln \omega_j} \quad (3.7)$$

where

$$\ln \omega_{der,j} = \frac{\ln \omega_{j+1} + \ln \omega_j}{2}. \quad (3.8)$$

3.2.4 Fitting

To understand the fitting process, a short introduction into the SciPy module `curve_fit` is needed. The function takes two arrays $[x_j]$, $[y_j]$ for the measured x and y data and a function $f(x, \theta)$ where θ is an arbitrary tuple of fit parameters to be optimized. The core of the `curve_fit` function is an iteration with changing guesses θ_l for the parameters. In every iteration step l the sum of squared differences between the measured y data and the results of the fit function is calculated.^[19]

$$\sum_j (y_j - f(x_j, \theta_l))^2 \quad (3.9)$$

This sum is minimized by very powerful optimization methods, that are not important for this discussion. The function f does not need to be a well-defined function in the mathematical sense that assigns to each element of the domain exactly one value of the codomain. Instead, it can be any logic function that takes an array $[x_j]$ and returns an array $f([x_j], \theta)$ of the same shape based on the parameters θ . The function is even allowed to return different values for the same x value at different positions j . Important is solely how close the results are to the respective y_j values.^[19]

Combined Fit

The measured data of ε' and ε'' could be combined into complex values and fitted using the function (3.1), which returns complex values as well. However, it is desired to take advantage of the better performance of ε'_{der} as discussed in 3.2.1. Therefore, two separated fit functions were used. For ε'' , the imaginary part can simply be calculated by applying `numpy.imag()` to the function (3.1). ε'_{der} could be handled similarly using `numpy.real()` and `numpy.gradient()`. However, in order to maintain the shape of the array, `numpy.gradient()` uses approximations at the edges. And while there are probably workarounds using the forward differentiation method (3.7) on transformed ω values, it is simpler to use the analytical calculation (3.3).

These two functions are then combined into one fit function with combined parameters using some logic: The respective x and y arrays are concatenated into two combined arrays $\mathbf{x}_{all}$ and $\mathbf{y}_{all}$, whereby $\mathbf{x}_{all}$ additionally stores the information, which function to use at which index.

Other Details

After trying out different combinations of various models, it was decided to fit three Cole/Cole processes in the measurements for 20-PEI and two for 43-PEI. For every run, a starting initial guess was manually defined that served as the initial guess θ_0 for the fit of the first spectrum. Every subsequent spectrum was fitted with the optimal parameters of the previous fit. Due to this method, one needs to be wary of “drift” effects. The fitting was performed in a log-log regime to ensure equal weighting across multiple orders of magnitude. Broad constraints were applied to the fit parameters: $\Delta\varepsilon > 0$, $\alpha \in [0, 1]$ according to their definitions^[3] and for each k ω_{Pk} was constrained to a broad range to ensure consistency of the labeling throughout the entire measurement. The boundaries for ω_P were chosen large enough to prevent any “push” effects. Since hundreds of fits were performed per measurement run, the procedure was fully automated. In case a fit fails to converge or produces an unreasonable result, fallbacks were implemented that retry the fit with the run’s starting initial guess. If the fit is still not satisfactory, it is not included in the final result. To automatically rate the fit, a scoring system was used that considers the coefficient of determination (R^2) and the χ^2 value as well as the parameter covariances. High accuracy of the rating system is not claimed, but it sufficiently filters out bad fits.

Result

Using this approach, temporal evolutions of the fit parameters were obtained for every measurement and run, giving insight into the kinetics of mobility, relaxation strength and broadening of the observed relaxation processes. The area under the fit function for ε'_{der} can also be used to get a more accurate time evolution of $\varepsilon'(\omega_0) - \varepsilon'(\omega \rightarrow \infty)$. The latter gives insight into characteristic times of CO₂ uptake dynamics.

4 Results and Discussion

All measurements followed the general form illustrated by figure 4.1. Three relaxation processes, numbered in order of their respective mean frequencies, were observed:

- (i) low-frequency: Only partially within the measurement window or completely outside it. It was fitted with a whole Cole/Cole term to correct for its contribution if needed. The fit parameters do not carry meaningful information as only the edge of the process was inside the frequency window, if anything.
- (ii) mid-frequency: Shows a distinct peak, usually with both shoulders visible, allowing for a detailed insight into intensity, broadening and mean frequency.
- (iii) high-frequency: is often strongly overlapped with Process (ii) and may include several sub-processes that cannot be individually resolved. Nevertheless, changes in combined relaxation strength still provide useful information.

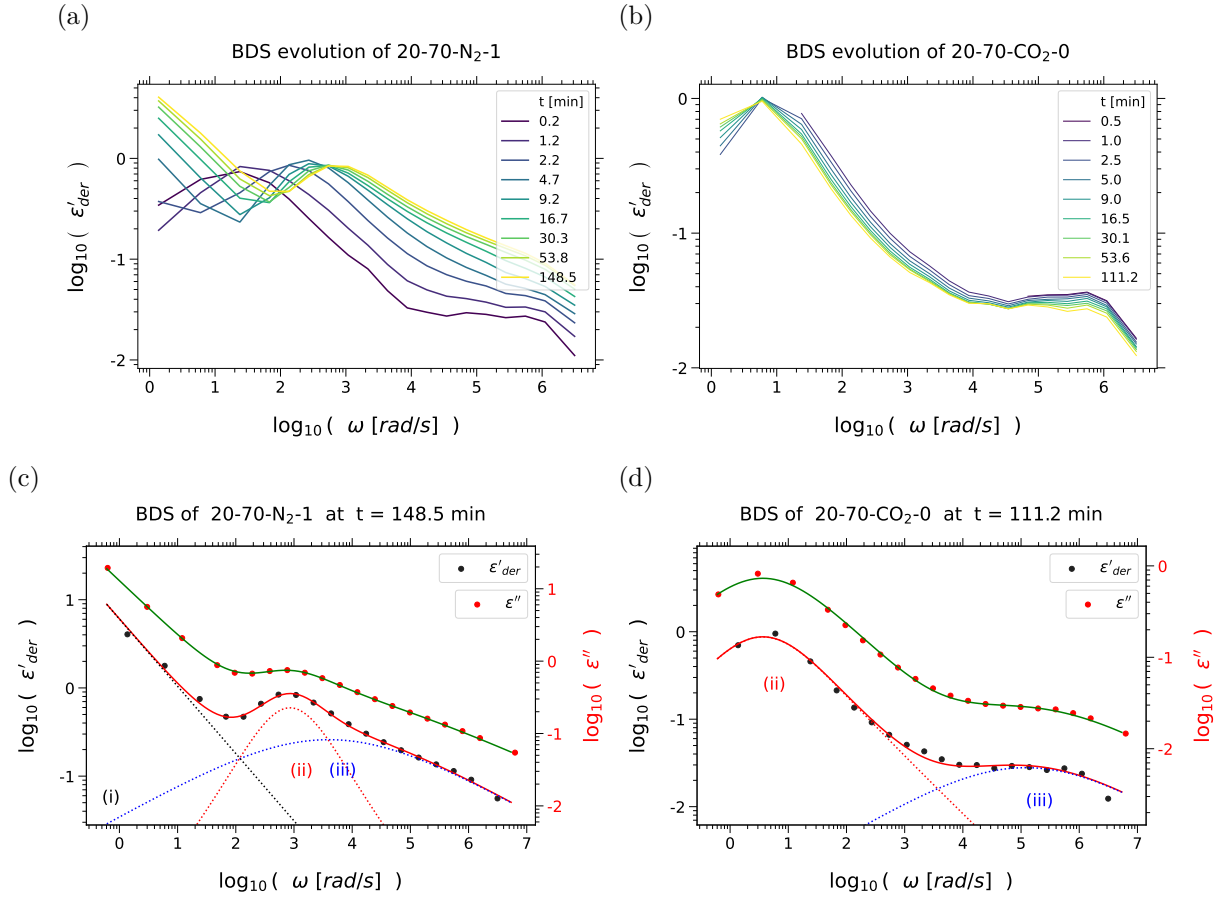


Fig. 4.1 Example spectra from the 70°C measurement. Overview of (a) desorption and (b) absorption kinetics as temporal evolution of the ϵ'_{der} spectrum. The measurements of other temperatures behave similarly and are shown in the appendix A. (c, d) Illustrations of combined fit for ϵ'' and ϵ'_{der} with three Cole/Cole terms indexed (i), (ii) and (iii) with rising mean frequencies. ϵ'' shifted upwards for clarity.

Similar relaxation processes were found in dielectric analysis of thin-film and bulk PEI by Tress et al.^[2], although shifted to faster frequencies by about a decade (comparing their steady state spectrum at 66.9°C with the present 70°C measurement). To reference their low-, mid- and high-frequency terms, (I), (II) and (III) will be used, respectively.

Upon switching from N₂ to CO₂, an instantaneous and strong transformation of the spectrum was observed. Remarkably, the shape of the spectrum remained almost static after this initial change. This implies either a rapid inversion of the desorption process or the existence of a mask obfuscating access to the molecular scale, as further interpreted in section 4.2.

This general picture is consistent across temperatures, though the dynamics slowed down significantly at lower temperatures. The following sections will quantify the results.

4.1 Desorption

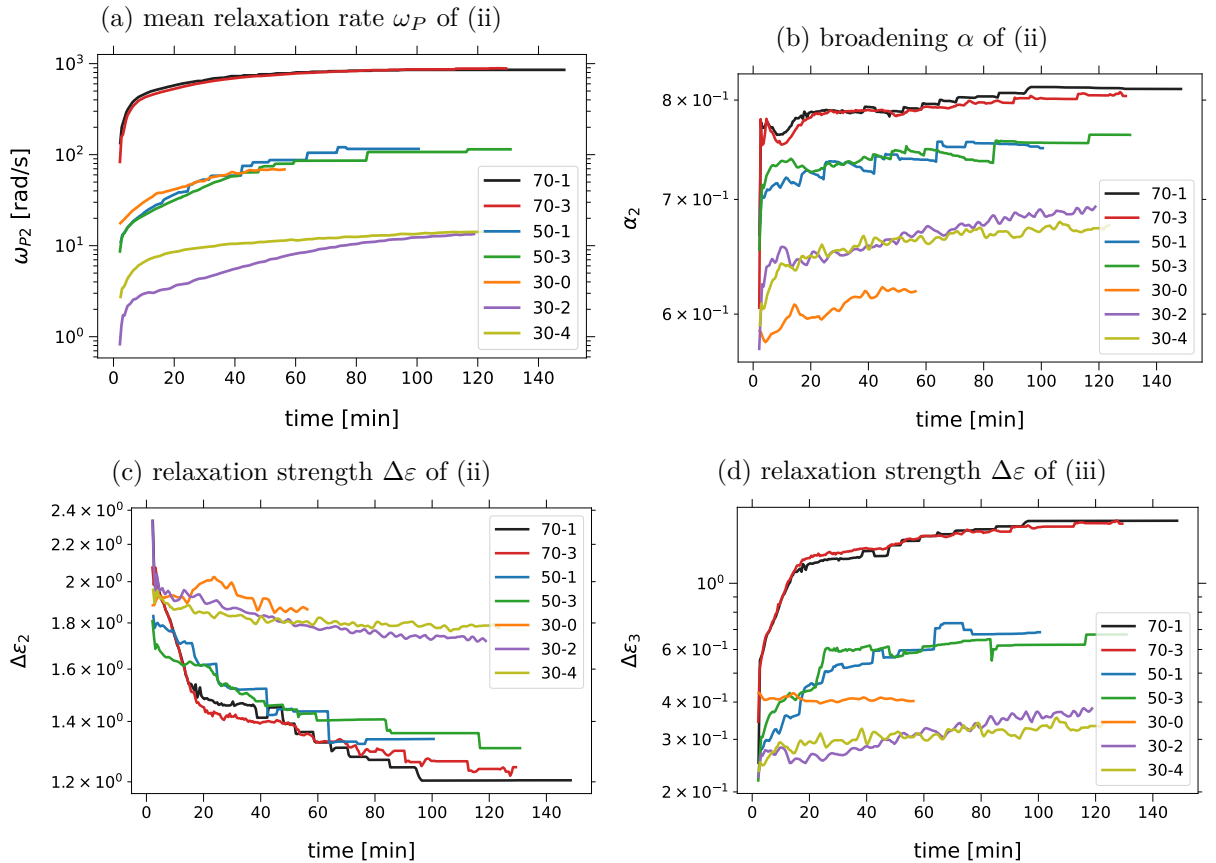


Fig. 4.2 Temporal evolutions of the most notable fit parameters of the desorption runs with 20-PEI. Legend encoding: TT - j with TT being the temperature in °C and j the run number.

Figure 4.2 shows the time evolution of key fit parameters for processes (ii) and (iii) during desorption. Desorption means exposure of CO₂ saturated samples to a N₂ atmosphere.

Between repeated runs, excellent reproducibility was found, with the exception of the 30°C measurements.

While the exact molecular origin of the relaxations remains uncertain, the anticorrelated evolution of relaxation strengths $\Delta\epsilon_2$ and $\Delta\epsilon_3$ during desorption aligns with the hypothesis by Tress et al., which suggests a dynamic exchange between free amine groups and ammonium carbamate species. More specifically, they associate the slower of the two relaxations with branch-scale motion, whereas the high-frequency peak corresponds to local mobility of amine groups or other segments. The large broadening of the latter, found in both studies, would correlate with the variety of amine types in the polymer.

The formation of charged ammonium carbamate and $-\text{NH}_3^+$ species upon CO_2 sorption supposedly results in strong ionic inter-branch connections. These were found to shield the affected dipoles from the outer electric field and stiffen the motion of the polymer branches. So they don't contribute directly to the polarization but reveal in significant change in the mobility of the relaxation (II). Furthermore, a more heterogeneous relaxation is expected with more species of the charged type, due to interatomic interactions and long range coulombic interactions.^[2]

Assuming correspondence of (ii) and (iii) and (II) and (III) of Tress et al., respectively, excellent compliance was found with the silica-induced PEI. (ii) shows a decade shift toward higher frequencies (fig. 4.2a) and a trend towards more homogeneity (fig. 4.2b) during desorption. Despite its complexity, intensity of (iii) offers a reasonable proxy for the overall amine concentration. It indicates desorption of CO_2 and an increase in the number of amine groups. At higher temperatures, the increase in intensity of (iii) after about 100 minutes is significantly stronger, which is consistent with faster desorption (fig. 4.2d).

Anomalies in the 30°C runs

Fig. 4.2a shows a surprisingly high mean relaxation rate ω_P of (ii) for the first run at 30°C (immediately after sample preparation, curve 30-0), implying significantly higher chain mobility. The evolution was similar to the 50°C runs. Since 30-0 was a desorption run directly following the preconditioning step, it was not expected to develop at all. At least, the higher mobility in run 0 could be explained with very little desorption of chemical bounds in the later runs due to the low temperature. Macroscopic desorption measurements (see section 2.2.2) support this, showing a slow but persistent desorption process at 30°C.

A notable feature in the second 30°C desorption run (fig. 4.2a, curve 30-2) is a dip in the mobility trajectory. It is known that the balloon was freshly prepared for 30-2, while 30-4 was conducted with a balloon that was stored for some time. This suggests the sensitivity

of the measurement to a temperature gradient introduced by the intake gas (as explained in section 3.1.4, Gas Temperature). As the balloon temperature matches room temperature in run 2, the mobility approaches the 4th run. The effect is expected to be more pronounced at lower measurement temperatures: The intake gas temperature difference $\Delta T := T_R - T_i$ (with room temperature T_R and initial gas temperature T_i) is constant, but should have a stronger effect on the temperature inside the capacitor for measurement temperatures closer to T_R . Indeed, a similar, much less significant dip shows in run 70-2, though balloon history could not be reconstructed for this measurement.

4.2 Absorption

As previously mentioned, CO₂ exposure results in an immediate transformation of the dielectric spectrum. The sharp change upon gas switch suggests a strong influence of the atmospheric composition. Indeed, CO₂ is more likely to form partial charges across the molecule than N₂. This may cause shielding of amine groups via physical interactions with CO₂, reducing their contribution to the polarization. This interpretation would further validate the rapid gas diffusion across the whole Molecular Basket system. At temperatures 30°C and 50°C, the slower dynamics push the main relaxation peaks outside the measured frequency window. Figure 4.3 shows parameter trends in the 70°C measurement that roughly mirror the desorption, but the mentioned problem obfuscates the results. While the trends look plausible, the actual values partially deviate. For a better comparison, the steady state fit parameter values are plotted in appendix D.

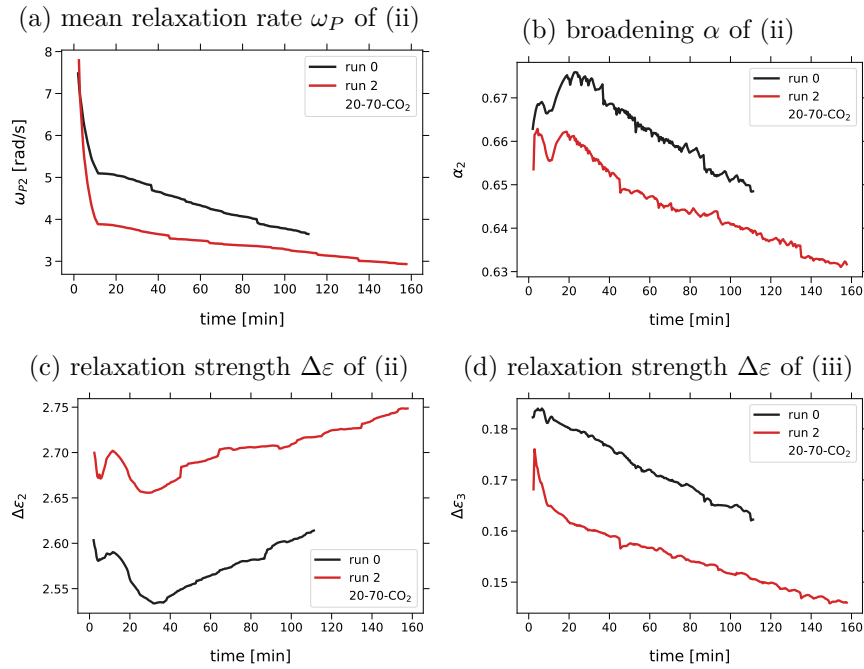


Fig. 4.3 Temporal evolutions of the most notable fit parameters of the absorption runs with 20-PEI. Legend encoding as specified in section 3.1.3.

4.3 Exponential Relaxation Characterization

The time evolutions of the Cole/Cole parameters in Figure 4.2 exhibit exponential-like behavior (fig. 4.4). Furthermore, the superposition of a very fast (τ_1) and a slower (τ_2) kinetic process is observed. This resembles trends seen in macroscopic CO₂ uptake measurements, where a fast physisorption step is followed by slower chemisorption. However, it remains unclear why physisorption would influence dielectric properties, further indicating atmosphere-material interactions. Additionally, further research is needed on the influence of gas alone or physisorbed gas in mesoporous systems on the permittivity. Control measurements on the empty mesoporous silica suggested slightly higher permittivity by about 1 % under CO₂ than under N₂ (see appendix E). The faster kinetic process could also just be an artifact of the time needed to fully exchange the atmosphere in the measurement cell. The changing gas concentration in the first few minutes could introduce an additional gradient influencing absorption speed.

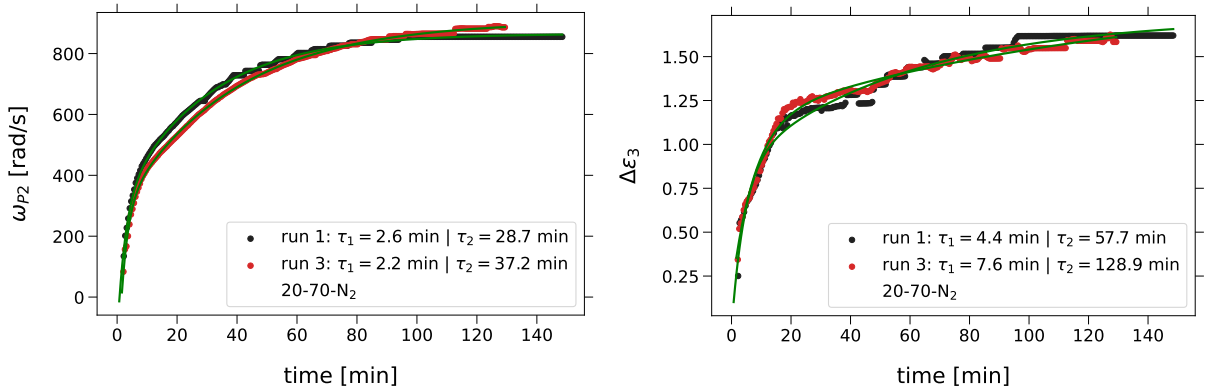


Fig. 4.4 Examples of the fitted exponential time-evolution of the spectral fit parameters. Fit models motivated and specified in section 3.2.1. All fits plotted in appendix B.

To analyze the kinetics, the parameters $\Delta\epsilon_2$, $\Delta\epsilon_3$, and ω_{p2} were fitted using either the double-exponential model (equation 3.6) or, – if good convergence or two distinct, meaningful characteristic times could not be reached – the single-exponential model (equation 3.5). In the latter case, the first 10–30 minutes were excluded to isolate τ_2 . These fits are attached as appendix B.

While analogous fits are not possible for the absorption runs, the slowed down polymer dynamics also affect the general area of the ϵ'_{der} spectrum. A numerical pseudo-real part $\langle\epsilon'\rangle_{\omega_1}$ was calculated from the ϵ'_{der} spectrum as area under the fit between the angular frequencies ω_1 and 10^9 . Higher frequencies can be ignored as the loss is practically zero. Although this approach risks peak shifts across the lower cutoff, it produced clean and noise-free curves that correlate well with double-exponential fits (appendix C). This was found to be the best method to extract kinetic information from the absorption measurements.

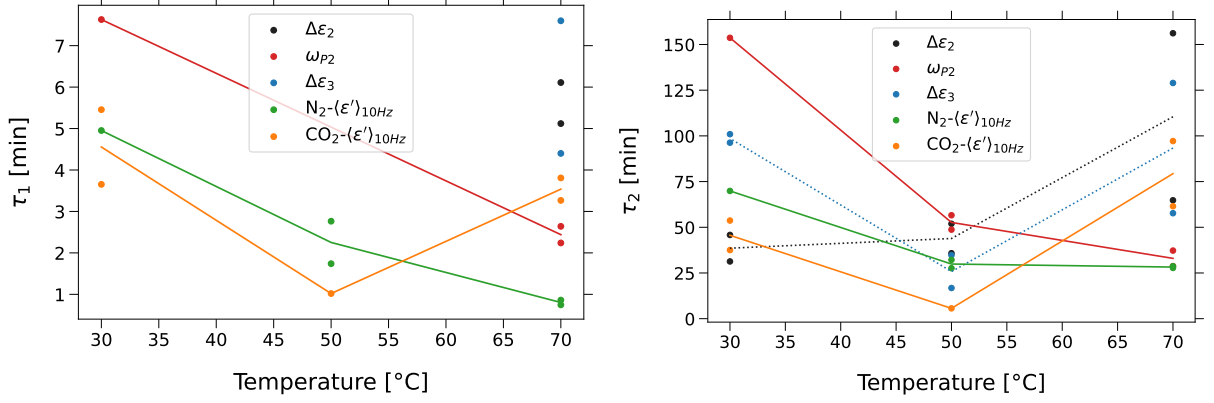


Fig. 4.5 Characteristic relaxation times of exponential fits of temporal evolutions of the fit parameters and the fit area. Temporal evolution showed superposition of two simple exponential functions. Shorter one on the left if resolvable and longer one on the right. Applied models specified in section 3.2.1.

Figure 4.5 shows that desorption kinetics accelerate with temperature, in line with previously mentioned macroscopic measurements. While τ -values derived from $\Delta\epsilon_2$ and $\Delta\epsilon_3$ are rough due to noise, their order of magnitude is consistent with the other values. Very prominent are the extremely fast kinetics found in absorption at 50°C. This could be an indication of interesting dynamics happening at 50°C as postulated by T. Beger et al.^[1], but the method to obtain this value was not robust enough to make any conclusions. More specifically, they suggested the combined influence of (a) polymer mobility, affecting gas diffusion, and (b) thermal energy of the binding sites, affecting the lifespan of the bound state, which corresponds to a kinetic resistance. (a) was proposed to be more prominent at lower temperatures, while (b) at higher temperatures, resulting in an optimal temperature around 50°C. It should be noted that the effect occurred at an atmosphere-like CO_2 concentration, which was not given in the present measurements and should affect gas diffusion considerations.

4.4 43-PEI

Additional measurements using 43-PEI, a system with fully filled silica pores and outside PEI coating^[7], revealed significant electrical conductivity (fig. 4.6), similar to prior measurements of pure bulk PEI. This indicates that the PEI network is continuous enough to allow ionic conduction between the capacitor plates, affecting the dielectric spectra. The strong polarization overshadows the relaxation process (ii), but exponential fits are still possible. Hence, further analysis is possible on this data, but different models and considerations are needed, exceeding the scope of this study. The calculations and plots as presented for 20-PEI exist for the 43-PEI measurements as well and are available on GitHub^[17], if desired.

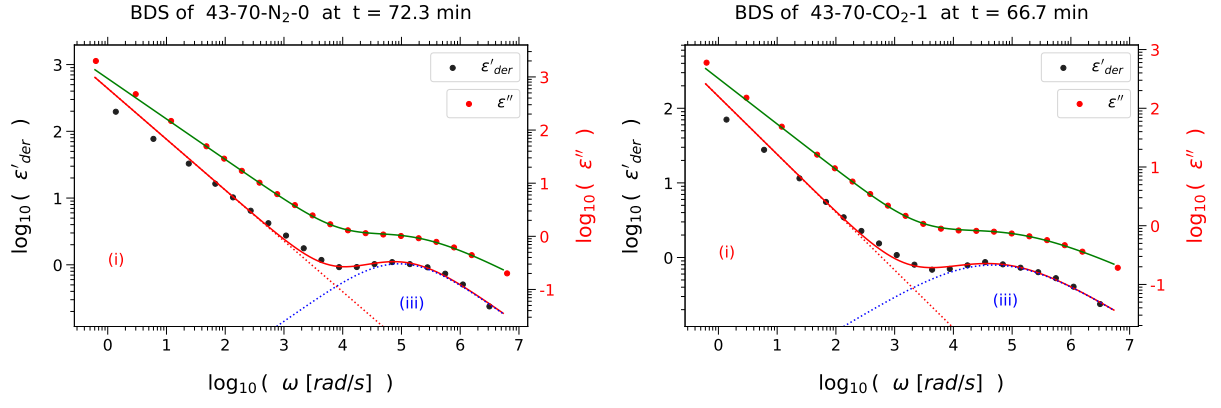


Fig. 4.6 Example spectra from the 70°C measurement. Illustrations of combined fit for ϵ'' and ϵ'_{der} with two Cole/Cole terms indexed (i) and (iii). ϵ'' shifted upwards for clarity. Process (ii) probably overlaid by strong polarization (i).

5 Conclusion and Outlook

We explored the molecular dynamics of polyethylenimine within a mesoporous silica framework using dielectric spectroscopy. The results demonstrated the potential of BDS to better understand Molecular Basket sorbents on a molecular level. Three relaxation processes were found, two of them attributed to relaxation of polymer chain motion and to the motion of amine groups. Screening the dielectric spectra over time revealed an exponential temporal evolution of amine concentration upon exposure to a N_2 atmosphere. Additionally, exponential evolution of the chain mobility was found, indicating cross-linking between polymer branches. These were explained by interactions of the ionic reaction products of the CO_2 chemisorption. Characteristic times could be extracted, revealing slower desorption at lower temperatures and surprising results for absorption at $50^\circ C$. While the measurements were not robust enough to make conclusive statements about the latter, it offers a root for possible further research. For better accuracy of the temporal evolution measurements, a time interpolation method was developed. This was needed to approximate dielectric spectra of a fixed time for all frequencies. The interpolation method can be applied to time-dependent BDS measurements in general.

This was the first of a potential series of measurements on Molecular Basket sorbents. Different pore diameters, volumes and fillings could be compared and the concept applied to different CO_2 sorptive materials within the mesoporous silica. Sensitivity of the measurement method to intake gas temperature was shown, which needs to be considered for future measurements. Rapid transformation of the spectrum was found upon CO_2 exposure. The present time resolution did not suffice to understand what happens. Future measurements should be done at better time resolution by concentrating on a smaller frequency window, preferably around the process (ii) associated with polymer chain motion. If further experiments are desired on this matter, I want to recommend a changing set of measured frequencies over time. When changing the intake gas, only few frequencies should be measured to capture fast changes. As the measurement progresses towards a steady state, frequencies should be added, to get a better spectral resolution. The code of the automated data import and interpolation should work for a changing set of frequencies and is freely available on GitHub. Note that because of the interpolation, the set of frequencies should only be monotonically growing or shrinking. Removing a frequency and re-adding it later leads to an interpolation across the whole intermission.

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Appendix

List of Annexes

A Spectral Evolutions by Temperature and Gas	ii
B Temporal Parameter Evolution Fits	iii
C Temporal Area Evolution Fits	iv
D Steady State Fit Parameters	v
E Empty Silica Measurements	vi

A Spectral Evolutions by Temperature and Gas

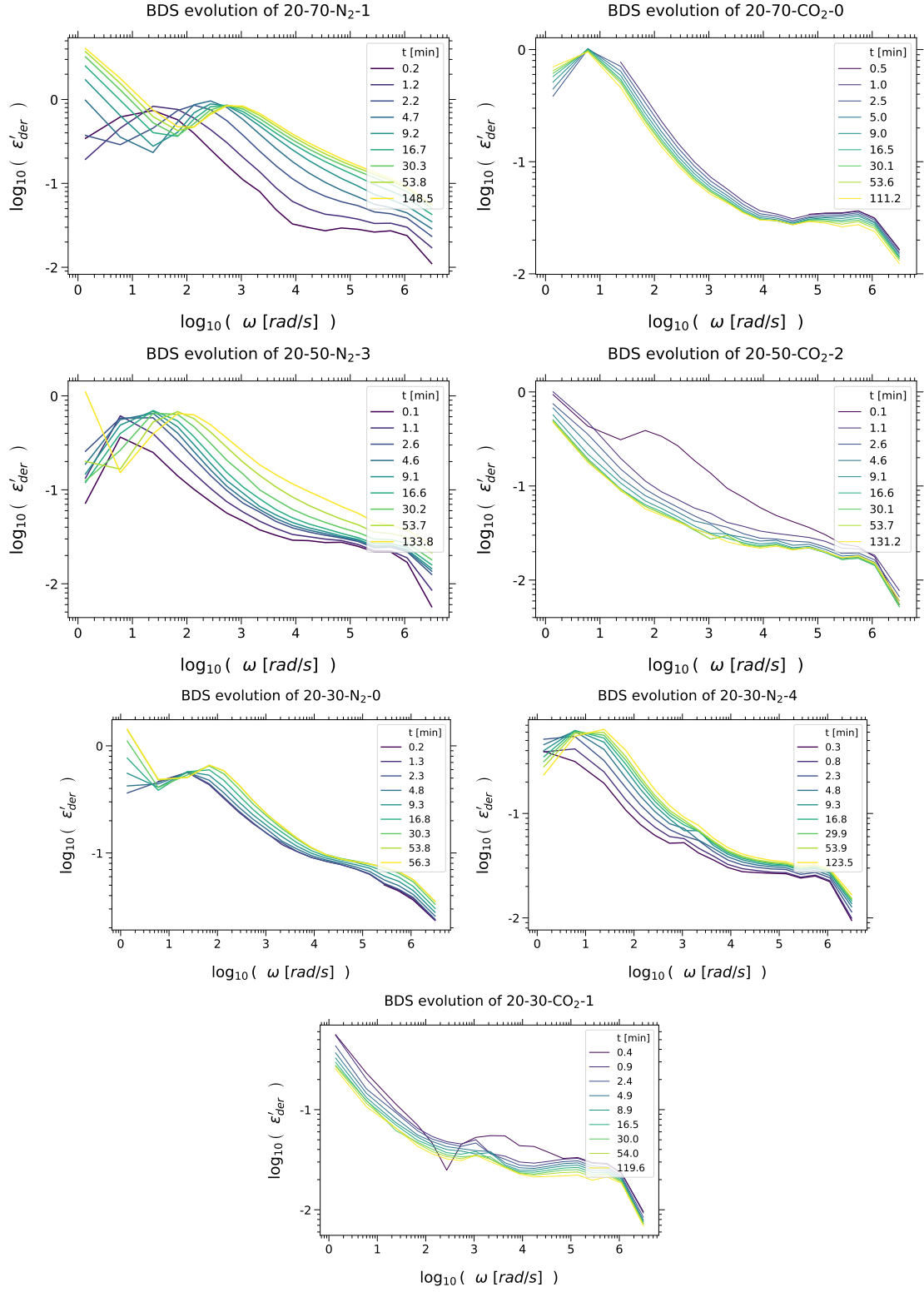


Fig. A.1 For every temperature and gas an overview of the evolution of the ϵ'_{der} spectrum for one example run. The different runs of one type were nearly identical, except for the desorption at 30°C. 30-0 deviates from the other two as discussed in results.

B Temporal Parameter Evolution Fits

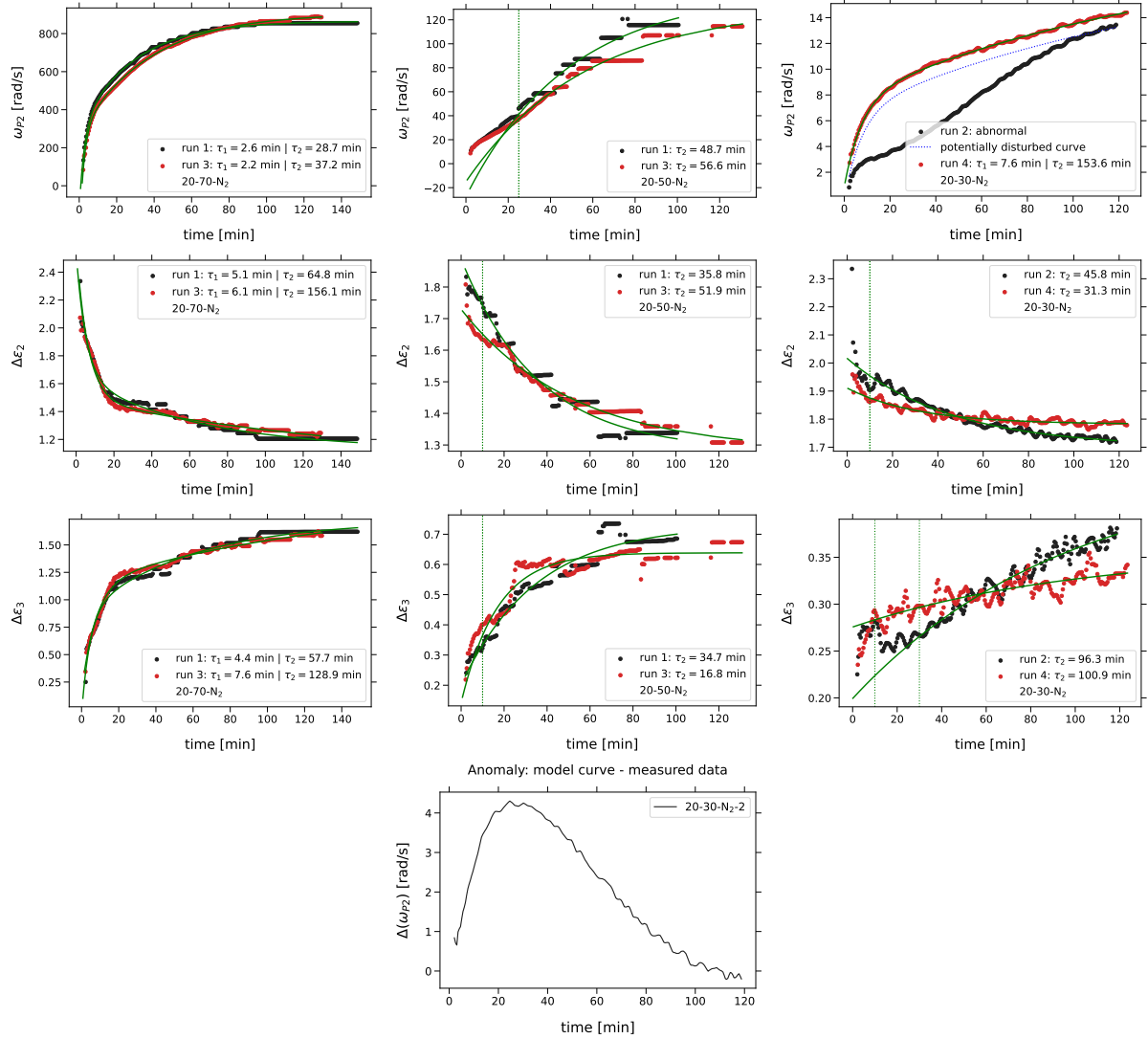


Fig. B.1 The Results section contains characteristic times for the evolution of the fit parameters ω_2 , $\Delta\epsilon_2$ and $\Delta\epsilon_3$. Those were obtained from the above fits. Used was either a simple exponential function or a sum of two as defined in Models, 3.2.1. In the former case, a section at the start was not included in the fit, indicated by the green dotted line. The fit may describe the start quite well nonetheless.

C Temporal Area Evolution Fits

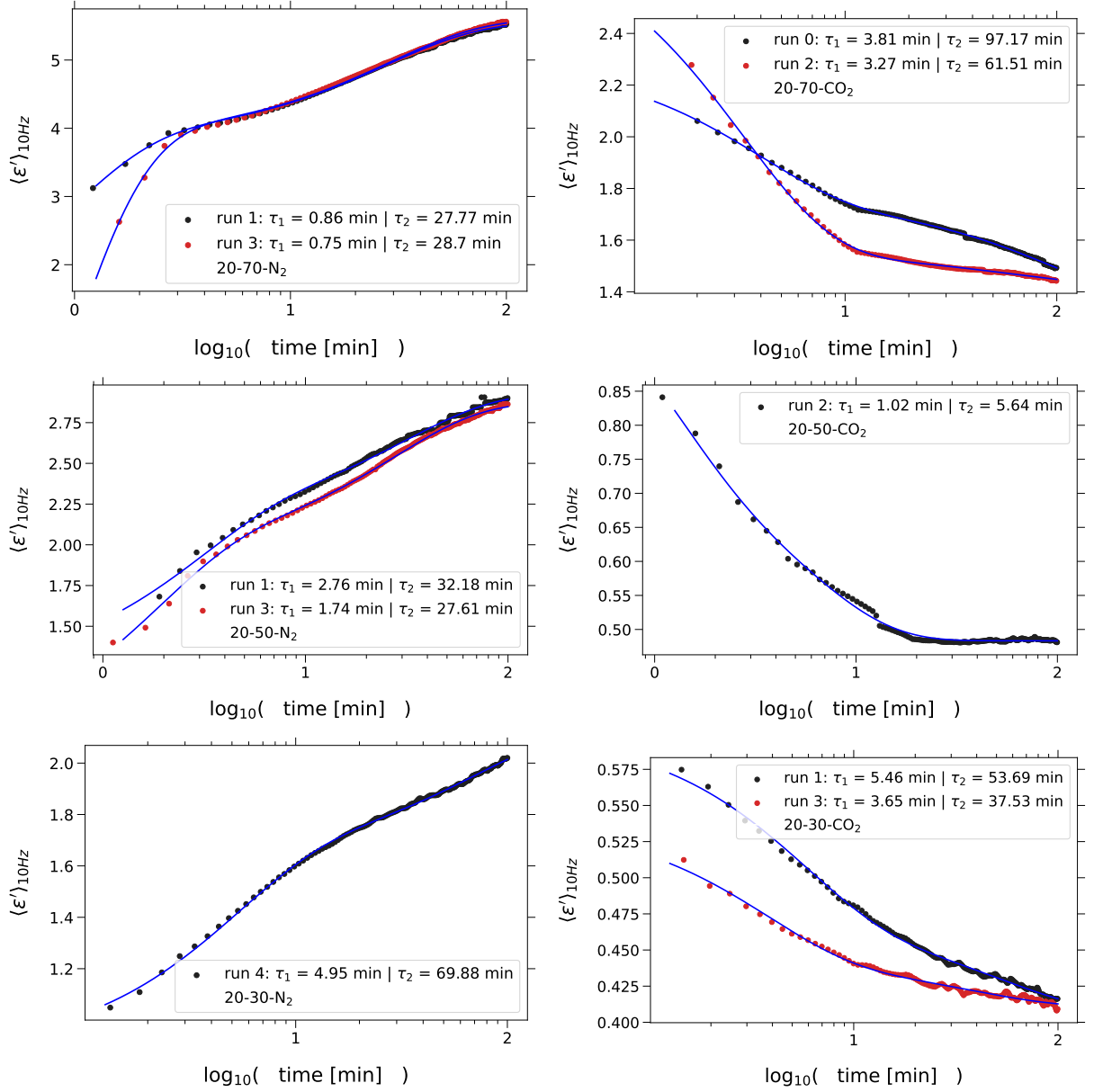


Fig. C.1 More accurate calculation of the real part of the permittivity ϵ' plotted over time. Calculated as a numerical area under the ϵ'_{der} spectrum between $\omega_1 = 10 \text{ rad/s}$ and $\omega_\infty = 10^9 \text{ rad/s}$. Above that, the fit is essentially zero and contributions above 10^6 rad/s should be minimal. Fitted as sum of two simple exponentials with relaxation times τ_1 and τ_2 .

D Steady State Fit Parameters

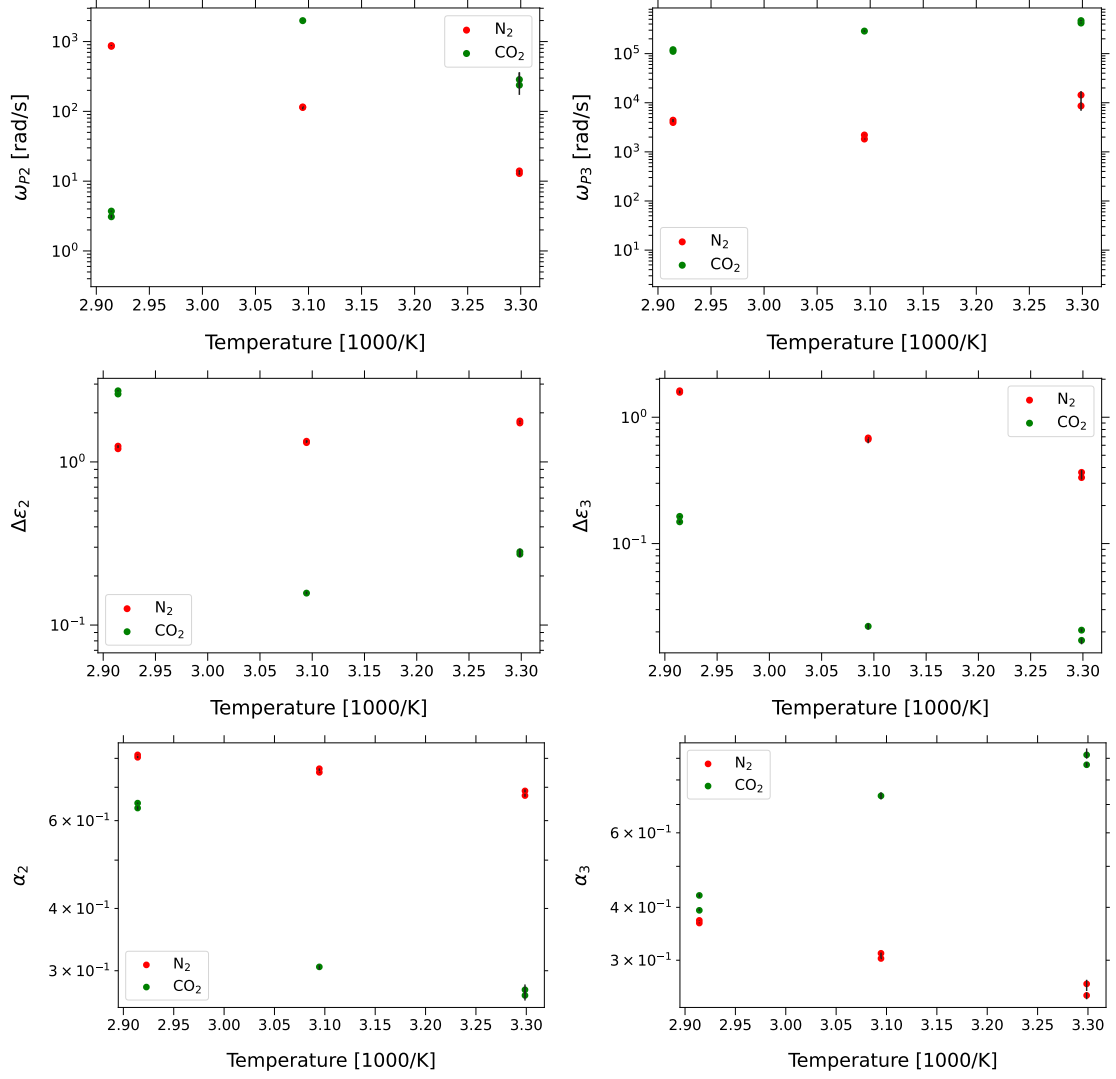


Fig. D.1 Average value of the respective spectral fit parameter after $t = 100$ min. Black lines represent the deviation of the values after $t = 100$ min. Those are mostly smaller than the symbol size. Points of different runs may overlap and appear as a single point.

E Empty Silica Measurements

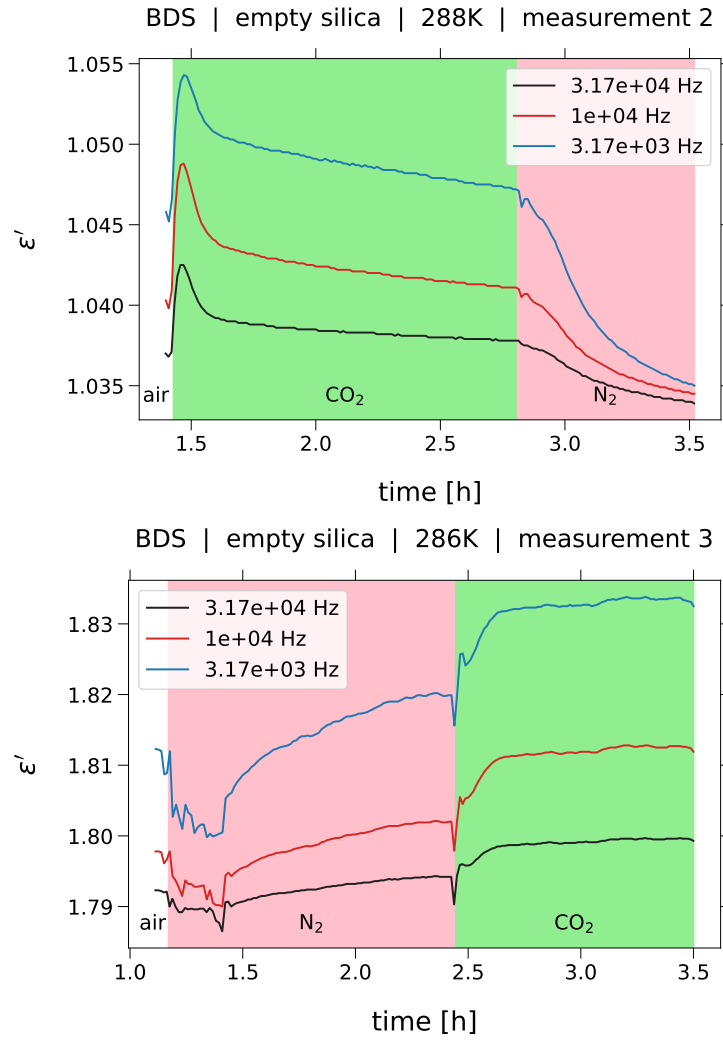


Fig. E.1 Baseline measurements with the empty silica. Imply influence of the gas alone, or physisorped gas. Spectral analysis might give further insights.